PSO — resultados CEC

Benchmark MAREA · paper OE1 · 71 funciones CEC2014/2017/2022 · 51 corridas por funcion

Protocolo

Valores = **error al optimo**, f(x) - f*. Menor es mejor; 0 es el optimo global. Dominio [-100,100] en todas las funciones. Poblacion N=80, iteraciones T=MaxFES/N. Presupuestos 5.000 / 50.000 / 500.000 evaluaciones. Las 30 funciones de CEC2014 y las 29 de CEC2017 corren en D=10 y D=50; las 12 de CEC2022 solo llegan a D=20, asi que en la columna D50 corren en **D=20** (marcado en la tabla). Todos los resultados vienen del mismo entorno del Mac mini (python 3.11.9, numpy 2.3.3, opfunu 1.0.4).

Resumen — posicion entre los 11 algoritmos

dimension	presupuesto	rango medio de Friedman	posicion de 11	funciones donde es el mejor
D10	5k FES	3.68	3º	5 de 71
D10	50k FES	4.25	3º	8 de 71
D10	500k FES	5.75	6º	4 de 71
D50	5k FES	4.89	5º	0 de 71
D50	50k FES	5.68	6º	1 de 71
D50	500k FES	7.92	10º	1 de 71

PSO — estadisticas por funcion, D = 10

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	10	2.02e+05	2.91e+07	1.00e+06	1.93e+06	4.03e+06	8.83e+02	2.85e+06	1.72e+05	2.94e+05	5.49e+05	2.96e+02	2.85e+06	7.68e+04	2.77e+05	5.54e+05
F22014	2014	10	5.45e+04	1.39e+06	3.22e+05	3.77e+05	2.44e+05	9.41e+00	1.18e+04	6.22e+03	6.15e+03	4.68e+03	3.95e-01	1.18e+04	4.27e+02	4.51e+03	5.42e+03
F32014	2014	10	3.20e+03	5.26e+04	1.94e+04	2.11e+04	1.20e+04	6.06e+00	7.35e+03	1.44e+03	2.01e+03	1.98e+03	1.09e-09	6.93e-03	1.22e-05	2.58e-04	9.94e-04
F42014	2014	10	1.17e+00	1.45e+02	3.48e+01	4.01e+01	3.02e+01	2.39e-01	1.45e+02	3.48e+01	3.49e+01	2.71e+01	1.51e-02	1.45e+02	3.48e+01	3.48e+01	2.71e+01
F52014	2014	10	7.72e+00	2.07e+01	2.06e+01	2.03e+01	1.80e+00	5.68e-14	2.04e+01	2.01e+01	1.97e+01	2.82e+00	0.00e+00	2.01e+01	2.00e+01	1.96e+01	2.80e+00
F62014	2014	10	5.82e-01	8.47e+00	3.16e+00	3.29e+00	1.67e+00	3.53e-05	8.38e+00	2.47e+00	2.65e+00	1.74e+00	3.53e-05	8.38e+00	2.39e+00	2.61e+00	1.74e+00
F72014	2014	10	5.36e-01	2.22e+01	4.39e+00	5.25e+00	5.22e+00	4.18e-02	2.20e+01	3.81e+00	4.47e+00	5.08e+00	4.18e-02	2.20e+01	3.78e+00	4.46e+00	5.08e+00
F82014	2014	10	6.20e+00	3.98e+01	1.86e+01	2.00e+01	6.51e+00	1.99e+00	2.10e+01	6.97e+00	8.16e+00	4.29e+00	9.95e-01	2.10e+01	6.97e+00	8.00e+00	4.33e+00
F92014	2014	10	6.45e+00	5.33e+01	2.71e+01	2.87e+01	1.10e+01	5.97e+00	4.08e+01	1.59e+01	1.77e+01	8.21e+00	5.97e+00	4.08e+01	1.39e+01	1.64e+01	8.41e+00
F102014	2014	10	3.63e+01	1.04e+03	4.74e+02	4.43e+02	2.46e+02	6.77e+00	7.50e+02	2.24e+02	2.66e+02	1.78e+02	6.77e+00	7.50e+02	2.24e+02	2.64e+02	1.81e+02
F112014	2014	10	1.42e+02	1.75e+03	9.02e+02	9.61e+02	3.83e+02	1.14e+02	1.26e+03	5.69e+02	5.41e+02	2.74e+02	7.14e+01	1.26e+03	5.25e+02	5.17e+02	2.69e+02
F122014	2014	10	3.82e-01	2.22e+00	1.34e+00	1.35e+00	4.56e-01	2.14e-02	4.80e-01	1.33e-01	1.64e-01	1.11e-01	2.10e-02	3.85e-01	9.35e-02	1.08e-01	7.65e-02
F132014	2014	10	3.08e-01	7.49e-01	4.81e-01	4.76e-01	1.01e-01	8.61e-02	5.55e-01	2.69e-01	2.74e-01	9.92e-02	8.61e-02	4.80e-01	2.57e-01	2.63e-01	9.97e-02
F142014	2014	10	1.93e-01	1.25e+00	4.61e-01	5.98e-01	3.28e-01	7.66e-02	9.89e-01	2.79e-01	4.60e-01	3.16e-01	5.64e-02	9.81e-01	1.94e-01	4.09e-01	3.36e-01
F152014	2014	10	1.38e+00	5.89e+00	3.54e+00	3.55e+00	8.35e-01	4.54e-01	1.93e+00	8.99e-01	9.07e-01	3.57e-01	4.54e-01	1.75e+00	7.98e-01	8.07e-01	2.61e-01
F162014	2014	10	2.68e+00	3.75e+00	3.32e+00	3.29e+00	2.93e-01	1.60e+00	3.48e+00	2.58e+00	2.58e+00	4.93e-01	7.33e-01	3.43e+00	2.52e+00	2.37e+00	5.53e-01
F172014	2014	10	2.05e+03	8.87e+04	2.69e+04	3.31e+04	2.00e+04	2.38e+02	7.41e+04	6.55e+03	1.28e+04	1.59e+04	2.43e+01	6.14e+04	1.52e+03	8.23e+03	1.27e+04
F182014	2014	10	1.77e+03	1.03e+05	2.29e+04	3.10e+04	2.24e+04	2.47e+00	1.03e+05	5.95e+03	1.46e+04	1.96e+04	1.76e-01	1.03e+05	2.16e+03	1.29e+04	2.00e+04
F192014	2014	10	5.74e+00	1.88e+02	1.82e+01	3.76e+01	4.35e+01	1.99e+00	1.80e+02	8.55e+00	2.29e+01	3.69e+01	1.77e+00	1.79e+02	7.33e+00	1.92e+01	3.47e+01
F202014	2014	10	5.88e+03	8.77e+06	3.92e+04	5.47e+05	2.04e+06	1.25e+03	8.57e+06	2.25e+04	5.28e+05	2.03e+06	3.50e+00	8.57e+06	1.14e+04	5.22e+05	2.03e+06
F212014	2014	10	7.12e+02	5.80e+05	1.15e+04	3.77e+04	1.11e+05	2.12e+01	5.63e+05	5.38e+02	1.59e+04	7.86e+04	8.04e-01	5.63e+05	4.65e+02	1.54e+04	7.86e+04
F222014	2014	10	2.52e+01	3.77e+03	2.45e+02	3.02e+02	5.26e+02	2.04e+01	3.77e+03	2.38e+02	2.71e+02	5.33e+02	2.02e+01	3.77e+03	2.37e+02	2.66e+02	5.34e+02
F232014	2014	10	2.37e+02	2.78e+02	2.42e+02	2.44e+02	1.02e+01	2.00e+02	2.72e+02	2.42e+02	2.41e+02	8.78e+00	2.00e+02	2.72e+02	2.42e+02	2.41e+02	8.78e+00
F242014	2014	10	1.15e+02	1.82e+02	1.35e+02	1.36e+02	1.24e+01	1.05e+02	1.50e+02	1.23e+02	1.25e+02	8.86e+00	1.05e+02	1.45e+02	1.22e+02	1.24e+02	8.53e+00
F252014	2014	10	1.61e+02	2.07e+02	2.01e+02	2.00e+02	6.76e+00	1.53e+02	2.06e+02	2.00e+02	2.00e+02	7.69e+00	1.53e+02	2.06e+02	2.00e+02	1.99e+02	7.69e+00
F262014	2014	10	1.00e+02	1.01e+02	1.01e+02	1.01e+02	1.83e-01	1.00e+02	1.01e+02	1.00e+02	1.00e+02	1.59e-01	1.00e+02	1.01e+02	1.00e+02	1.00e+02	1.53e-01
F272014	2014	10	6.64e+00	5.84e+02	4.02e+02	4.05e+02	9.11e+01	3.05e+00	5.75e+02	4.01e+02	4.00e+02	8.97e+01	2.59e+00	5.75e+02	4.00e+02	4.00e+02	8.98e+01
F282014	2014	10	3.94e+02	7.07e+02	4.84e+02	4.89e+02	7.49e+01	3.94e+02	6.92e+02	4.84e+02	4.86e+02	7.12e+01	3.94e+02	6.92e+02	4.84e+02	4.86e+02	7.10e+01
F292014	2014	10	4.50e+06	1.54e+07	5.80e+06	6.55e+06	2.17e+06	4.50e+06	1.54e+07	5.80e+06	6.52e+06	2.19e+06	4.50e+06	1.54e+07	5.80e+06	6.52e+06	2.19e+06
F302014	2014	10	2.63e+03	3.10e+07	1.10e+05	1.63e+06	5.18e+06	2.51e+03	2.60e+07	5.31e+04	1.49e+06	4.64e+06	2.45e+03	2.60e+07	5.31e+04	1.49e+06	4.64e+06
F12017	2017	10	1.12e+05	1.35e+09	6.80e+05	1.54e+08	3.95e+08	4.34e+01	1.35e+09	5.53e+03	1.53e+08	3.95e+08	4.14e+01	1.35e+09	5.53e+03	1.53e+08	3.95e+08
F22017	2017	10	2.06e+00	6.63e+02	3.40e+01	8.50e+01	1.51e+02	0.00e+00	6.46e+02	0.00e+00	1.27e+01	9.04e+01	0.00e+00	6.46e+02	0.00e+00	1.27e+01	9.04e+01
F32017	2017	10	1.75e+00	8.06e+01	9.25e+00	2.50e+01	2.87e+01	1.52e-02	7.49e+01	9.22e+00	1.38e+01	2.14e+01	8.70e-04	7.49e+01	9.22e+00	1.25e+01	2.02e+01
F42017	2017	10	4.49e+01	1.99e+03	7.36e+01	4.35e+02	5.46e+02	3.98e+00	1.97e+03	1.69e+01	3.79e+02	5.53e+02	3.98e+00	1.97e+03	1.39e+01	3.79e+02	5.53e+02
F52017	2017	10	1.75e-07	6.39e-04	9.71e-06	6.39e-05	1.23e-04	0.00e+00	6.36e-04	7.97e-09	2.32e-05	9.46e-05	0.00e+00	1.76e-04	2.58e-10	1.08e-05	3.60e-05
F62017	2017	10	7.42e+01	3.96e+04	9.05e+03	7.85e+03	9.49e+03	1.19e+01	3.95e+04	2.85e+01	6.72e+03	9.45e+03	8.95e+00	3.95e+04	2.85e+01	6.72e+03	9.45e+03
F72017	2017	10	2.00e+01	8.42e+01	4.74e+01	4.75e+01	1.42e+01	1.00e+01	3.80e+01	2.20e+01	2.22e+01	7.49e+00	1.00e+01	3.60e+01	2.00e+01	2.09e+01	7.30e+00
F82017	2017	10	6.45e-04	1.77e+00	1.23e-02	2.40e-01	3.79e-01	0.00e+00	1.77e+00	0.00e+00	2.02e-01	3.59e-01	0.00e+00	1.77e+00	0.00e+00	2.02e-01	3.59e-01
F92017	2017	10	1.38e+02	1.85e+03	6.45e+02	7.26e+02	3.43e+02	1.18e+01	9.45e+02	3.89e+02	3.74e+02	2.17e+02	1.18e+01	9.45e+02	3.89e+02	3.68e+02	2.12e+02
F102017	2017	10	2.13e+02	5.94e+04	1.26e+04	1.79e+04	1.56e+04	1.13e+00	3.58e+04	1.40e+03	6.28e+03	9.03e+03	9.95e-01	3.58e+04	1.32e+03	6.04e+03	8.98e+03
F112017	2017	10	1.02e+04	7.85e+07	8.43e+04	5.65e+06	1.62e+07	2.35e+02	4.84e+07	1.64e+04	1.20e+06	6.94e+06	1.75e+02	4.84e+07	1.60e+04	1.20e+06	6.94e+06
F122017	2017	10	4.07e+03	6.27e+06	2.30e+05	3.31e+05	8.71e+05	1.59e+02	4.45e+05	1.54e+04	1.71e+05	2.05e+05	1.56e+02	4.44e+05	1.54e+04	1.71e+05	2.05e+05
F132017	2017	10	3.07e+02	4.57e+04	1.47e+04	1.72e+04	1.04e+04	2.92e+01	9.17e+03	5.67e+02	1.22e+03	1.75e+03	2.20e+01	9.14e+03	2.99e+01	3.29e+02	1.45e+03
F142017	2017	10	3.87e+03	1.92e+06	4.20e+04	1.52e+05	3.53e+05	2.95e+02	1.67e+06	1.47e+04	9.18e+04	3.03e+05	2.41e+00	1.67e+06	8.49e+03	8.76e+04	3.02e+05
F152017	2017	10	7.31e+00	1.45e+04	7.34e+02	1.47e+03	2.37e+03	6.96e-02	1.09e+03	4.08e+02	3.86e+02	3.05e+02	5.29e-02	1.09e+03	4.04e+02	3.73e+02	3.00e+02
F162017	2017	10	6.93e+01	2.30e+04	6.51e+02	1.25e+03	3.23e+03	2.11e+01	1.23e+03	4.60e+01	2.16e+02	2.71e+02	2.01e+01	1.23e+03	2.92e+01	1.88e+02	2.50e+02
F172017	2017	10	2.91e+03	7.85e+04	2.69e+04	3.11e+04	1.73e+04	7.14e+01	6.23e+04	5.07e+03	1.16e+04	1.54e+04	2.00e+01	6.01e+04	1.04e+03	8.42e+03	1.40e+04
F182017	2017	10	1.59e+03	7.71e+04	1.56e+04	1.80e+04	1.61e+04	1.48e+00	7.22e+04	3.27e+03	9.39e+03	1.41e+04	9.39e-01	7.20e+04	6.17e+00	7.14e+03	1.38e+04
F192017	2017	10	3.10e+01	3.49e+02	8.23e+01	1.15e+02	8.01e+01	1.42e+00	2.41e+02	3.68e+01	6.79e+01	6.78e+01	9.97e-01	2.40e+02	3.37e+01	6.49e+01	6.79e+01

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	10	1.08e+02	8.48e+02	5.85e+02	4.75e+02	2.24e+02	1.00e+02	8.47e+02	5.84e+02	4.63e+02	2.32e+02	1.00e+02	8.47e+02	5.84e+02	4.63e+02	2.32e+02
F212017	2017	10	1.04e+02	1.31e+02	1.12e+02	1.13e+02	6.15e+00	1.03e+02	1.31e+02	1.11e+02	1.11e+02	6.23e+00	1.03e+02	1.31e+02	1.11e+02	1.11e+02	6.20e+00
F222017	2017	10	1.30e+02	1.41e+03	7.65e+02	6.64e+02	3.82e+02	1.00e+02	1.41e+03	7.58e+02	6.18e+02	4.05e+02	1.00e+02	1.41e+03	7.58e+02	6.18e+02	4.05e+02
F232017	2017	10	1.20e+02	2.62e+03	1.27e+03	1.44e+03	8.07e+02	1.00e+02	2.51e+03	1.27e+03	1.41e+03	8.24e+02	1.00e+02	2.51e+03	1.27e+03	1.41e+03	8.24e+02
F242017	2017	10	4.79e+02	5.74e+02	4.84e+02	4.98e+02	2.65e+01	4.76e+02	5.69e+02	4.82e+02	4.94e+02	2.50e+01	4.76e+02	5.69e+02	4.82e+02	4.94e+02	2.50e+01
F252017	2017	10	6.04e-01	5.61e+02	4.69e+02	4.63e+02	9.69e+01	4.55e-13	5.61e+02	4.69e+02	4.63e+02	9.69e+01	4.55e-13	5.61e+02	4.69e+02	4.63e+02	9.69e+01
F262017	2017	10	3.95e+02	4.88e+02	4.06e+02	4.10e+02	1.52e+01	3.95e+02	4.88e+02	4.06e+02	4.10e+02	1.53e+01	3.95e+02	4.88e+02	4.06e+02	4.09e+02	1.53e+01
F272017	2017	10	6.15e+00	5.04e+02	4.05e+02	3.80e+02	1.40e+02	4.55e-13	5.04e+02	3.99e+02	3.73e+02	1.48e+02	0.00e+00	5.04e+02	3.99e+02	3.73e+02	1.48e+02
F282017	2017	10	1.07e+03	9.94e+04	6.49e+03	1.17e+04	1.56e+04	9.89e+02	9.94e+04	4.77e+03	8.54e+03	1.50e+04	9.89e+02	9.94e+04	3.49e+03	8.38e+03	1.50e+04
F292017	2017	10	7.45e+03	1.92e+07	3.43e+04	1.60e+06	4.01e+06	7.67e+02	1.92e+07	2.11e+04	1.46e+06	3.98e+06	2.57e+02	1.92e+07	2.11e+04	1.46e+06	3.98e+06
F12022	2022	10	2.26e+00	8.89e+01	1.22e+01	1.83e+01	1.70e+01	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22022	2022	10	1.31e+00	1.04e+02	1.95e+01	3.67e+01	3.44e+01	3.85e-01	1.04e+02	1.95e+01	3.55e+01	3.34e+01	3.85e-01	1.04e+02	1.95e+01	3.55e+01	3.34e+01
F32022	2022	10	7.29e-06	8.30e-02	7.39e-05	1.57e-02	2.33e-02	0.00e+00	8.29e-02	0.00e+00	1.54e-02	2.32e-02	0.00e+00	8.29e-02	0.00e+00	1.54e-02	2.32e-02
F42022	2022	10	2.08e+01	1.14e+02	6.00e+01	6.02e+01	2.14e+01	1.00e+01	7.60e+01	3.40e+01	3.63e+01	1.52e+01	1.00e+01	5.80e+01	2.60e+01	2.83e+01	1.03e+01
F52022	2022	10	5.86e-04	5.20e+00	9.11e-02	3.26e-01	7.79e-01	0.00e+00	2.06e+00	3.41e-13	1.90e-01	3.38e-01	0.00e+00	2.06e+00	2.27e-13	1.90e-01	3.38e-01
F62022	2022	10	9.37e+03	1.10e+05	4.92e+04	5.35e+04	2.44e+04	3.74e+03	1.04e+05	2.63e+04	3.22e+04	2.28e+04	4.20e+01	1.04e+05	1.75e+04	2.48e+04	2.34e+04
F72022	2022	10	2.27e+01	1.59e+02	4.72e+01	6.52e+01	4.29e+01	1.05e-03	1.31e+02	2.20e+01	4.09e+01	4.02e+01	8.61e-05	1.31e+02	2.15e+01	4.05e+01	4.00e+01
F82022	2022	10	2.37e+01	2.67e+03	6.05e+01	2.86e+02	6.31e+02	2.30e-01	2.66e+03	2.13e+01	9.29e+01	3.83e+02	7.42e-02	2.66e+03	2.06e+01	9.23e+01	3.83e+02
F92022	2022	10	1.15e+02	7.09e+02	3.80e+02	3.94e+02	6.68e+01	1.14e+02	7.09e+02	3.80e+02	3.90e+02	6.61e+01	1.14e+02	7.09e+02	3.80e+02	3.90e+02	6.61e+01
F102022	2022	10	2.21e+02	5.37e+02	2.32e+02	2.74e+02	6.68e+01	2.21e+02	5.36e+02	2.32e+02	2.72e+02	6.56e+01	2.21e+02	5.36e+02	2.29e+02	2.70e+02	6.53e+01
F112022	2022	10	4.51e-01	9.00e+02	2.72e+01	1.94e+02	3.42e+02	4.55e-13	9.00e+02	2.66e+01	1.93e+02	3.42e+02	0.00e+00	9.00e+02	2.66e+01	1.93e+02	3.42e+02
F122022	2022	10	1.67e+02	2.30e+02	1.71e+02	1.73e+02	8.75e+00	1.67e+02	2.30e+02	1.71e+02	1.73e+02	8.76e+00	1.67e+02	2.30e+02	1.71e+02	1.73e+02	8.76e+00

PSO — estadisticas por funcion, D = 50

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	50	1.51e+08	1.59e+09	4.19e+08	4.95e+08	2.87e+08	2.47e+07	1.59e+09	2.62e+08	3.60e+08	3.03e+08	1.81e+07	1.59e+09	2.58e+08	3.56e+08	3.06e+08
F22014	2014	50	1.99e+10	8.76e+10	4.35e+10	4.29e+10	1.31e+10	9.43e+09	6.79e+10	3.30e+10	3.29e+10	1.33e+10	9.43e+09	6.79e+10	3.30e+10	3.29e+10	1.33e+10
F32014	2014	50	1.97e+05	3.86e+05	3.00e+05	2.96e+05	3.88e+04	3.43e+04	1.81e+05	8.85e+04	8.77e+04	3.09e+04	5.77e+01	4.76e+04	7.50e+03	1.22e+04	1.28e+04
F42014	2014	50	1.08e+03	1.18e+04	3.77e+03	4.58e+03	2.82e+03	2.69e+02	1.10e+04	2.58e+03	3.27e+03	2.55e+03	2.69e+02	1.10e+04	2.58e+03	3.27e+03	2.55e+03
F52014	2014	50	2.12e+01	2.13e+01	2.13e+01	2.13e+01	3.92e-02	2.06e+01	2.13e+01	2.11e+01	2.11e+01	1.15e-01	2.00e+01	2.11e+01	2.02e+01	2.03e+01	3.57e-01
F62014	2014	50	4.35e+01	5.95e+01	5.23e+01	5.22e+01	3.49e+00	3.34e+01	5.29e+01	4.31e+01	4.26e+01	4.23e+00	3.34e+01	5.29e+01	4.30e+01	4.25e+01	4.25e+00
F72014	2014	50	1.70e+02	7.68e+02	3.89e+02	3.95e+02	1.27e+02	6.50e+01	6.01e+02	3.02e+02	2.97e+02	1.30e+02	6.50e+01	6.01e+02	3.02e+02	2.97e+02	1.30e+02
F82014	2014	50	3.57e+02	5.40e+02	4.64e+02	4.58e+02	5.26e+01	1.28e+02	3.30e+02	2.31e+02	2.36e+02	5.24e+01	1.28e+02	3.30e+02	2.31e+02	2.36e+02	5.24e+01
F92014	2014	50	4.96e+02	7.63e+02	6.00e+02	6.04e+02	6.48e+01	1.98e+02	5.18e+02	3.20e+02	3.20e+02	7.44e+01	1.98e+02	5.18e+02	3.20e+02	3.20e+02	7.45e+01
F102014	2014	50	8.51e+03	1.31e+04	1.05e+04	1.06e+04	1.14e+03	3.45e+03	8.35e+03	5.57e+03	5.59e+03	1.05e+03	3.45e+03	8.35e+03	5.57e+03	5.59e+03	1.05e+03
F112014	2014	50	9.79e+03	1.59e+04	1.29e+04	1.30e+04	1.29e+03	3.81e+03	8.91e+03	6.80e+03	6.83e+03	9.71e+02	3.81e+03	8.90e+03	6.79e+03	6.80e+03	9.62e+02
F122014	2014	50	2.95e+00	5.70e+00	4.55e+00	4.49e+00	5.90e-01	2.75e-01	2.67e+00	6.78e-01	7.95e-01	4.86e-01	1.79e-01	8.27e-01	4.56e-01	4.40e-01	1.29e-01
F132014	2014	50	2.42e+00	5.73e+00	4.06e+00	4.00e+00	7.54e-01	7.25e-01	5.34e+00	3.67e+00	3.26e+00	1.32e+00	7.25e-01	5.34e+00	3.67e+00	3.26e+00	1.32e+00
F142014	2014	50	3.69e+01	2.05e+02	9.27e+01	1.03e+02	4.08e+01	1.23e+00	1.95e+02	7.16e+01	7.90e+01	4.44e+01	1.14e+00	1.95e+02	7.16e+01	7.90e+01	4.44e+01
F152014	2014	50	5.54e+03	2.35e+06	2.66e+05	4.57e+05	4.79e+05	5.53e+01	1.33e+06	8.48e+04	2.05e+05	3.17e+05	5.20e+01	1.33e+06	8.48e+04	2.05e+05	3.17e+05
F162014	2014	50	2.21e+01	2.36e+01	2.30e+01	2.30e+01	3.04e-01	2.03e+01	2.25e+01	2.13e+01	2.13e+01	5.65e-01	1.92e+01	2.19e+01	2.04e+01	2.04e+01	5.71e-01
F172014	2014	50	5.21e+06	7.95e+07	1.62e+07	2.27e+07	1.69e+07	6.04e+05	2.31e+07	2.46e+06	3.94e+06	4.11e+06	1.35e+05	2.20e+07	8.29e+05	2.67e+06	4.00e+06
F182014	2014	50	4.89e+06	1.09e+09	2.87e+07	1.04e+08	2.13e+08	1.66e+05	9.10e+08	3.64e+05	4.34e+07	1.66e+08	1.62e+05	9.10e+08	3.54e+05	4.34e+07	1.66e+08
F192014	2014	50	1.70e+04	6.87e+07	3.94e+05	3.70e+06	1.18e+07	5.14e+01	6.24e+07	2.79e+02	1.85e+06	9.05e+06	5.14e+01	6.24e+07	2.18e+02	1.85e+06	9.05e+06
F202014	2014	50	8.55e+05	2.55e+11	1.78e+09	1.43e+10	4.36e+10	1.73e+05	1.35e+10	5.23e+05	8.90e+08	2.79e+09	1.23e+05	1.30e+10	4.25e+05	7.95e+08	2.52e+09
F212014	2014	50	4.99e+06	2.95e+08	2.79e+07	5.04e+07	5.84e+07	2.49e+05	2.31e+08	8.00e+06	2.83e+07	4.56e+07	4.71e+04	2.31e+08	6.62e+06	2.75e+07	4.58e+07
F222014	2014	50	4.98e+03	3.14e+08	1.01e+04	7.17e+06	4.41e+07	1.65e+03	3.10e+08	4.09e+03	6.08e+06	4.34e+07	8.30e+02	3.10e+08	3.14e+03	6.08e+06	4.34e+07
F232014	2014	50	3.14e+02	5.89e+02	3.82e+02	3.96e+02	6.27e+01	2.62e+02	5.36e+02	3.30e+02	3.45e+02	6.12e+01	2.62e+02	5.36e+02	3.29e+02	3.45e+02	6.15e+01
F242014	2014	50	3.71e+02	5.66e+02	4.55e+02	4.58e+02	4.09e+01	2.86e+02	5.07e+02	3.84e+02	3.88e+02	4.65e+01	2.86e+02	5.07e+02	3.84e+02	3.87e+02	4.70e+01
F252014	2014	50	2.28e+02	3.15e+02	2.56e+02	2.60e+02	1.76e+01	2.10e+02	2.77e+02	2.27e+02	2.30e+02	1.30e+01	2.07e+02	2.77e+02	2.26e+02	2.29e+02	1.35e+01
F262014	2014	50	1.05e+02	4.65e+02	2.24e+02	2.26e+02	1.05e+02	1.04e+02	4.54e+02	2.06e+02	2.03e+02	1.03e+02	1.04e+02	4.54e+02	2.05e+02	2.01e+02	1.04e+02
F272014	2014	50	7.75e+02	1.99e+03	1.81e+03	1.79e+03	1.76e+02	4.65e+02	1.93e+03	1.66e+03	1.65e+03	2.15e+02	4.57e+02	1.93e+03	1.65e+03	1.64e+03	2.16e+02
F282014	2014	50	2.33e+03	4.80e+03	3.57e+03	3.60e+03	6.57e+02	2.27e+03	4.75e+03	3.52e+03	3.52e+03	6.36e+02	2.27e+03	4.75e+03	3.43e+03	3.52e+03	6.36e+02
F292014	2014	50	7.40e+07	5.21e+08	1.75e+08	2.25e+08	1.25e+08	1.07e+05	5.21e+08	1.19e+08	1.56e+08	1.31e+08	3.92e+04	5.21e+08	1.19e+08	1.55e+08	1.31e+08
F302014	2014	50	4.53e+07	8.84e+09	1.32e+09	1.78e+09	1.71e+09	4.64e+06	9.09e+08	6.32e+07	1.57e+08	2.22e+08	4.62e+06	9.09e+08	6.09e+07	1.53e+08	2.23e+08
F12017	2017	50	9.94e+09	5.69e+10	2.92e+10	3.12e+10	1.17e+10	2.49e+09	4.95e+10	1.87e+10	2.13e+10	1.16e+10	2.49e+09	4.95e+10	1.87e+10	2.13e+10	1.16e+10
F22017	2017	50	6.75e+04	1.71e+05	1.02e+05	1.07e+05	2.31e+04	4.08e+01	5.58e+04	1.08e+04	1.26e+04	1.19e+04	5.68e-14	5.14e+04	8.72e+03	1.14e+04	1.14e+04
F32017	2017	50	8.46e+02	1.30e+04	3.92e+03	4.66e+03	2.73e+03	3.41e+02	9.47e+03	2.62e+03	3.19e+03	2.34e+03	3.40e+02	9.47e+03	2.62e+03	3.18e+03	2.35e+03
F42017	2017	50	1.46e+04	7.23e+04	3.32e+04	3.66e+04	1.47e+04	2.94e+03	6.69e+04	2.44e+04	2.80e+04	1.62e+04	2.94e+03	6.69e+04	2.43e+04	2.80e+04	1.62e+04
F52017	2017	50	7.65e-03	6.26e-02	2.32e-02	2.46e-02	1.25e-02	6.21e-05	7.84e-03	6.07e-04	1.15e-03	1.59e-03	9.28e-06	7.24e-03	4.08e-04	9.49e-04	1.35e-03
F62017	2017	50	4.82e+05	3.17e+06	1.46e+06	1.65e+06	6.90e+05	1.31e+05	3.06e+06	1.33e+06	1.36e+06	7.20e+05	9.54e+04	3.06e+06	1.33e+06	1.36e+06	7.24e+05
F72017	2017	50	8.79e+02	1.50e+03	1.20e+03	1.19e+03	1.37e+02	3.62e+02	9.98e+02	6.22e+02	6.37e+02	1.38e+02	3.62e+02	9.98e+02	6.16e+02	6.27e+02	1.35e+02
F82017	2017	50	2.22e+01	7.87e+01	4.64e+01	4.66e+01	1.37e+01	1.32e+01	6.27e+01	2.83e+01	3.06e+01	1.07e+01	1.32e+01	6.27e+01	2.81e+01	3.05e+01	1.07e+01
F92017	2017	50	1.12e+04	1.54e+04	1.36e+04	1.35e+04	1.03e+03	4.33e+03	8.94e+03	6.88e+03	7.02e+03	9.41e+02	4.32e+03	8.84e+03	6.81e+03	6.98e+03	9.37e+02
F102017	2017	50	4.07e+05	6.25e+09	5.44e+07	2.63e+08	8.85e+08	1.72e+05	2.59e+09	1.26e+07	1.79e+08	4.10e+08	1.67e+05	2.59e+09	1.26e+07	1.79e+08	4.10e+08
F112017	2017	50	1.77e+08	1.06e+10	1.31e+09	1.79e+09	1.76e+09	1.67e+06	1.06e+10	6.51e+08	1.29e+09	1.83e+09	1.67e+06	1.06e+10	6.48e+08	1.29e+09	1.83e+09
F122017	2017	50	4.14e+08	2.56e+10	5.44e+09	7.22e+09	6.48e+09	4.25e+05	2.46e+10	3.36e+09	5.62e+09	6.12e+09	4.24e+05	2.46e+10	3.34e+09	5.62e+09	6.12e+09
F132017	2017	50	8.30e+05	2.89e+07	4.66e+06	5.60e+06	5.08e+06	2.34e+05	2.73e+07	1.20e+06	2.47e+06	4.71e+06	1.23e+05	2.73e+07	6.94e+05	1.94e+06	4.77e+06
F142017	2017	50	1.13e+07	6.96e+09	1.08e+09	1.26e+09	1.52e+09	1.44e+05	6.93e+09	7.03e+08	1.12e+09	1.54e+09	1.35e+05	6.93e+09	7.01e+08	1.12e+09	1.54e+09
F152017	2017	50	2.28e+05	1.35e+09	5.55e+07	2.03e+08	3.10e+08	1.26e+05	1.29e+09	4.74e+07	1.73e+08	2.74e+08	1.26e+05	1.29e+09	4.74e+07	1.73e+08	2.74e+08
F162017	2017	50	2.98e+04	8.54e+07	1.33e+05	1.98e+06	1.19e+07	1.08e+03	1.79e+05	7.32e+04	8.03e+04	4.28e+04	4.89e+02	1.78e+05	7.11e+04	7.88e+04	4.27e+04
F172017	2017	50	1.30e+06	9.46e+07	1.25e+07	1.76e+07	1.77e+07	3.00e+05	2.09e+07	9.70e+05	2.63e+06	4.31e+06	3.16e+04	2.07e+07	4.67e+05	2.02e+06	3.85e+06
F182017	2017	50	6.86e+08	2.24e+13	7.63e+10	6.40e+11	3.18e+12	2.12e+05	2.15e+13	6.48e+10	6.01e+11	3.06e+12	2.12e+05	2.15e+13	6.48e+10	6.01e+11	3.06e+12
F192017	2017	50	1.66e+03	6.56e+03	3.84e+03	4.07e+03	1.25e+03	2.44e+02	3.40e+03	1.26e+03	1.33e+03	7.06e+02	2.37e+02	3.39e+03	1.26e+03	1.28e+03	6.77e+02

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	50	4.86e+03	5.66e+04	2.37e+04	2.29e+04	1.42e+04	4.36e+03	5.20e+04	1.29e+04	1.70e+04	1.23e+04	4.36e+03	5.20e+04	1.29e+04	1.70e+04	1.23e+04
F212017	2017	50	6.21e+02	1.15e+04	9.36e+02	1.75e+03	2.69e+03	5.39e+02	6.43e+03	8.66e+02	1.26e+03	1.24e+03	5.39e+02	6.43e+03	8.66e+02	1.26e+03	1.24e+03
F222017	2017	50	1.74e+04	5.04e+04	3.55e+04	3.55e+04	7.50e+03	9.77e+03	5.03e+04	3.09e+04	3.11e+04	8.82e+03	9.77e+03	5.03e+04	3.09e+04	3.10e+04	8.83e+03
F232017	2017	50	1.02e+04	3.69e+04	1.73e+04	1.89e+04	5.35e+03	2.21e+03	3.04e+04	1.66e+04	1.71e+04	6.55e+03	2.20e+03	3.04e+04	1.66e+04	1.71e+04	6.55e+03
F242017	2017	50	1.70e+03	1.01e+04	3.71e+03	4.03e+03	2.05e+03	9.89e+02	9.10e+03	2.33e+03	2.95e+03	1.88e+03	9.85e+02	9.10e+03	2.33e+03	2.95e+03	1.88e+03
F252017	2017	50	2.07e+03	6.59e+03	3.43e+03	3.62e+03	1.14e+03	2.07e+03	6.59e+03	3.43e+03	3.62e+03	1.14e+03	2.07e+03	6.59e+03	3.43e+03	3.62e+03	1.14e+03
F262017	2017	50	6.63e+02	1.37e+03	9.80e+02	1.00e+03	1.79e+02	6.47e+02	1.37e+03	9.61e+02	9.95e+02	1.81e+02	6.47e+02	1.37e+03	9.61e+02	9.95e+02	1.81e+02
F272017	2017	50	7.86e+02	4.04e+03	1.57e+03	1.64e+03	6.27e+02	6.33e+02	4.04e+03	1.56e+03	1.59e+03	6.45e+02	6.33e+02	4.04e+03	1.56e+03	1.59e+03	6.45e+02
F282017	2017	50	3.72e+07	6.24e+11	2.29e+09	1.81e+10	8.90e+10	5.19e+05	3.25e+11	5.88e+08	9.52e+09	4.74e+10	4.47e+05	3.25e+11	5.78e+08	9.52e+09	4.74e+10
F292017	2017	50	8.51e+08	1.24e+11	3.47e+10	4.62e+10	3.56e+10	1.66e+07	1.05e+11	2.35e+10	3.16e+10	2.85e+10	7.30e+06	1.05e+11	2.35e+10	3.16e+10	2.85e+10
F12022	2022	20	1.08e+03	7.57e+03	2.68e+03	3.17e+03	1.48e+03	0.00e+00	2.76e+03	5.68e-14	1.85e+02	5.38e+02	0.00e+00	2.76e+03	5.68e-14	1.85e+02	5.38e+02
F22022	2022	20	4.16e+01	5.55e+02	1.20e+02	1.42e+02	9.24e+01	3.23e+01	5.55e+02	9.19e+01	1.23e+02	9.28e+01	4.03e+00	5.55e+02	9.19e+01	1.18e+02	9.34e+01
F32022	2022	20	3.22e-03	4.71e-01	1.61e-01	1.72e-01	9.10e-02	1.14e-13	4.70e-01	1.57e-01	1.66e-01	9.33e-02	1.14e-13	4.70e-01	1.57e-01	1.66e-01	9.33e-02
F42022	2022	20	8.11e+01	3.17e+02	2.47e+02	2.42e+02	4.99e+01	5.80e+01	2.88e+02	1.40e+02	1.47e+02	5.60e+01	5.80e+01	2.88e+02	1.40e+02	1.45e+02	5.34e+01
F52022	2022	20	8.92e-01	1.51e+01	4.60e+00	4.39e+00	2.70e+00	6.33e-01	1.47e+01	4.06e+00	3.99e+00	2.64e+00	6.33e-01	1.47e+01	4.06e+00	3.99e+00	2.64e+00
F62022	2022	20	1.17e+05	1.06e+08	5.86e+05	7.93e+06	1.84e+07	3.69e+04	1.05e+08	1.09e+05	4.89e+06	1.65e+07	1.21e+04	1.05e+08	9.00e+04	4.81e+06	1.65e+07
F72022	2022	20	6.23e+01	1.55e+03	2.19e+02	2.92e+02	2.69e+02	2.32e+01	5.30e+02	2.93e+01	6.47e+01	9.92e+01	2.16e+01	5.30e+02	2.68e+01	5.44e+01	8.91e+01
F82022	2022	20	2.63e+02	1.09e+04	3.60e+03	3.87e+03	2.45e+03	2.85e+01	8.40e+03	8.28e+02	1.38e+03	1.80e+03	2.20e+01	8.40e+03	5.94e+02	1.21e+03	1.77e+03
F92022	2022	20	3.42e+02	1.12e+03	4.61e+02	5.08e+02	1.56e+02	3.36e+02	1.12e+03	4.61e+02	5.00e+02	1.54e+02	3.36e+02	1.12e+03	4.61e+02	5.00e+02	1.54e+02
F102022	2022	20	4.10e+02	4.57e+03	2.64e+03	2.18e+03	1.48e+03	3.99e+02	2.92e+03	1.62e+03	1.47e+03	8.98e+02	3.99e+02	2.92e+03	1.62e+03	1.46e+03	8.89e+02
F112022	2022	20	2.60e+01	1.96e+03	1.28e+02	6.64e+02	7.53e+02	2.32e+01	1.96e+03	1.28e+02	6.61e+02	7.55e+02	2.32e+01	1.96e+03	1.28e+02	6.61e+02	7.55e+02
F122022	2022	20	2.47e+02	5.52e+02	2.89e+02	3.04e+02	5.45e+01	2.47e+02	5.52e+02	2.89e+02	3.03e+02	5.46e+01	2.47e+02	5.52e+02	2.89e+02	3.02e+02	5.47e+01